

Amazon Sales Data Analysis and Business Intelligence Dashboard Using Power BI

1. Project Overview and Objective

Project Overview

This project focuses on analyzing Amazon sales data by performing data cleaning, transformation, and analysis using Microsoft Excel, followed by building an interactive Business Intelligence dashboard using Power BI. The project aims to convert raw transactional sales data into meaningful insights that help understand sales performance, profitability, customer segments, and regional trends.

Project Objective

The main objectives of this project are:

- To perform **data pre-processing and cleaning** using Excel.
 - To analysis **sales, profit, and discount patterns** across regions and product categories.
 - To create an **interactive Power BI dashboard** for business decision-making.
 - To demonstrate practical knowledge of **Excel, Power BI, and data analytics techniques**.
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2. Data Sources

Source Description

- Dataset Type: **Amazon Sales Dataset**
- Data Type: **Transactional Sales Data**
- Timeline: **2023 – 2025**

Domain E-Commerce

The dataset contains detailed information about customer orders, shipping details, product categories, sales values, discounts, and profits.

3. Problem Statement

The key problems addressed in this project are:

- To Analysis **overall sales and profit performance** across different regions.
 - To identify **top-performing product categories and sub-categories**.
 - To study the impact of **discounts on profit**.
 - To understand **customer segments and purchasing behavior**.
 - To support **business decision-making** through visual insights.
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4. Attribute (Column / Feature) Details

Attribute Name	Data Type	Description
Order ID	String	Unique identifier for each order
Order Date	Date	Date when the order was placed
Ship Date	Date	Date when the order was shipped
Customer Name	String	Name of the customer
Segment	Categorical	Customer segment (Consumer, Corporate, Home Office)
Country	String	Country where the order was placed
State	String	State of the customer
City	String	City of the customer
Region	Categorical	Sales region
Category	Categorical	Product category
Sub-Category	Categorical	Product sub-category
Product Name	String	Name of the product
Sales	Numeric (Decimal)	Total sales value

Attribute Name	Data Type	Description
Quantity	Numeric (Integer)	Quantity of products ordered
Discount	Numeric (Decimal)	Discount applied
Profit	Numeric (Decimal)	Profit earned from the order
Payment Mode	Categorical	Mode of payment

5. Tools & Technologies

Microsoft Excel -2024

- Data cleaning and transformation
- Handling missing values and duplicates
- Standardizing formats (date, currency, text)
- Creating calculated columns
- Pivot Tables for initial analysis

Power BI

- Data modelling (Fact & Dimension tables)
- DAX calculations
- Interactive dashboards and visualizations
- Filters, slicers, and drill-downs

6. Data Pre-Processing (Excel / Power Query)

Tasks Performed

Data Cleaning & Transformation

- Removed duplicate records.
- Handled missing values using appropriate techniques.
- Standardized date, currency, and text formats.
- Converted data types (text to number, date, percentage).
- Cleaned city, state, and country inconsistencies.

Filtering & Sorting

- Filtered irrelevant records.
- Sorted data by sales, profit, and date.

Calculated Fields

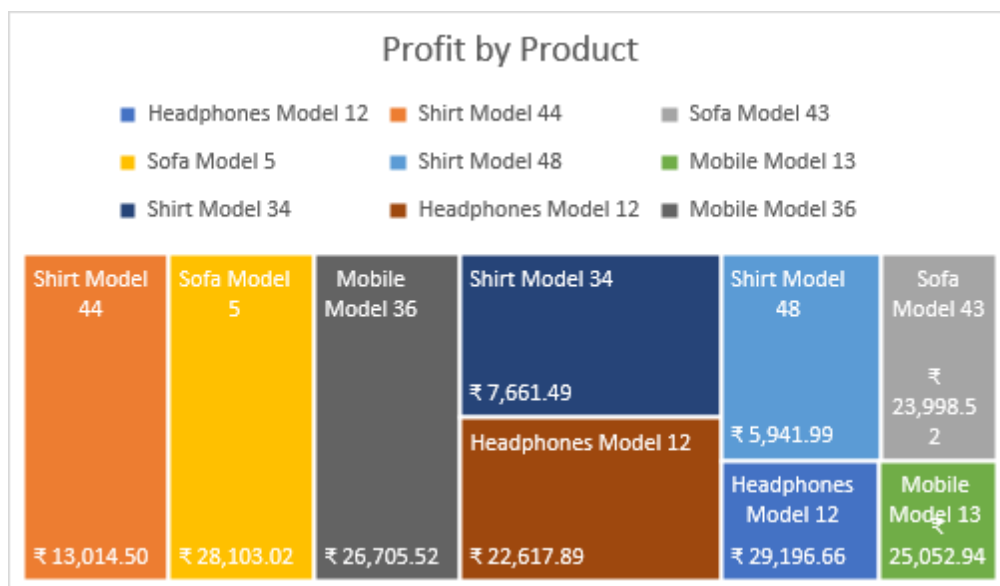
- Profit Margin
- Discount Percentage
- Shipping Duration (Ship Date – Order Date)

Pivot Tables

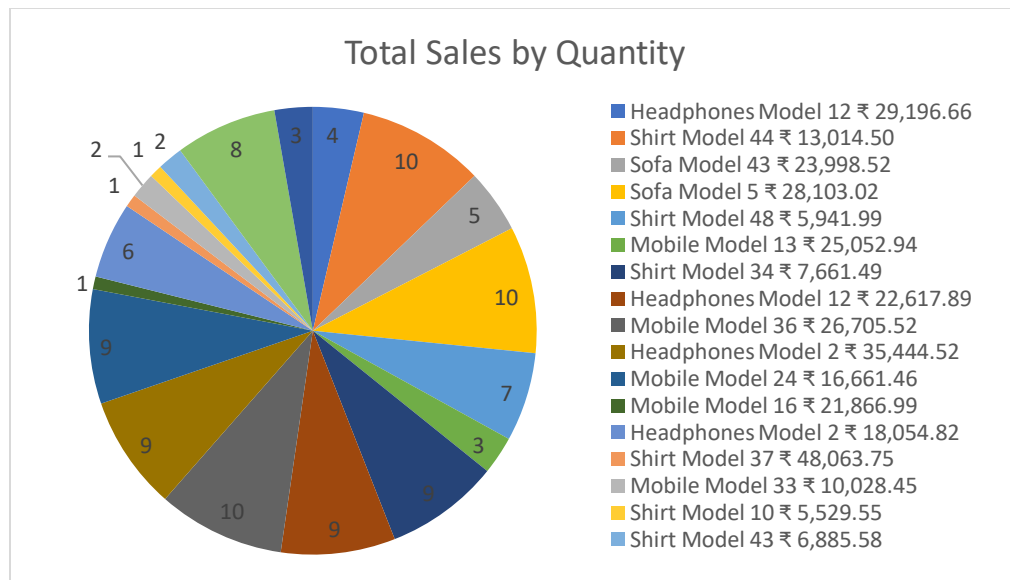
1. Sum of Profit using Quantity 2. Count of Sales, Profit and Discount

Row Labels	Sum of Quantity	Sum of Profit	Row Labels	Count of Sales	Sum of Profit	Count of Discount
ORD1000	39	7297.88	ORD1000	6	7297.88	6
2023-01-22	10	453.41	India	1	7416.43	1
2023-04-18	3	-2608.34	Headphones Model 13	1	7416.43	1
2023-06-20	9	1911.29	UK	3	578.5	3
2023-06-22	9	1911.29	Headphones Model 8	1	2153.77	1
John Doe	9	1911.29	Shirt Model 8	1	453.41	1
Home Office	9	1911.29	Sofa Model 23	1	-2028.68	1
USA	9	1911.29	USA	2	-697.05	2
Maharashtra	9	1911.29	Laptop Model 36	1	-2608.34	1
West	9	1911.29	Sofa Model 13	1	1911.29	1
Fashion	9	1911.29	ORD1001	13	60104.5	13
Sofa	9	1911.29	India	4	16117.7	4
Sofa Model 13	9	1911.29	Laptop Model 18	1	2177.97	1
16490.04	9	1911.29	Mobile Model 26	1	2831.98	1
22%	9	1911.29	Shirt Model 32	1	8680.08	1
COD	9	1911.29	Sofa Model 44	1	7427.67	1
London	9	1911.29				

- Profit by Product



- Total Sales by Quantity



7. Data Modelling (Fact & Dimension Tables)

Fact Table

- **Fact_Sales**
 - Order ID
 - Sales
 - Quantity
 - Discount
 - Profit

Dimension Tables

- **Dim_Customer** (Customer Name, Segment)
 - **Dim_Product** (Category, Sub-Category, Product Name)
 - **Dim_Location** (Country, State, City, Region)
 - **Dim_Date** (Order Date, Ship Date, Year, Month)
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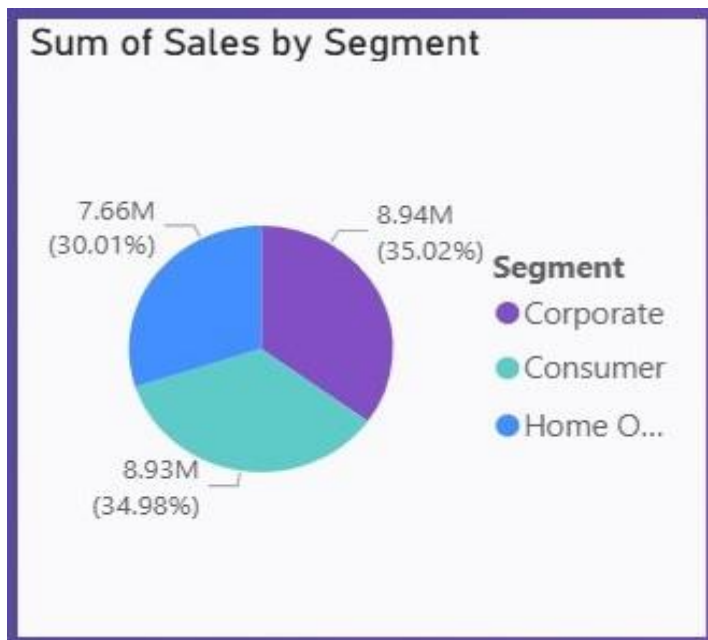
8. Analysis and (Power BI)

Dashboard Features

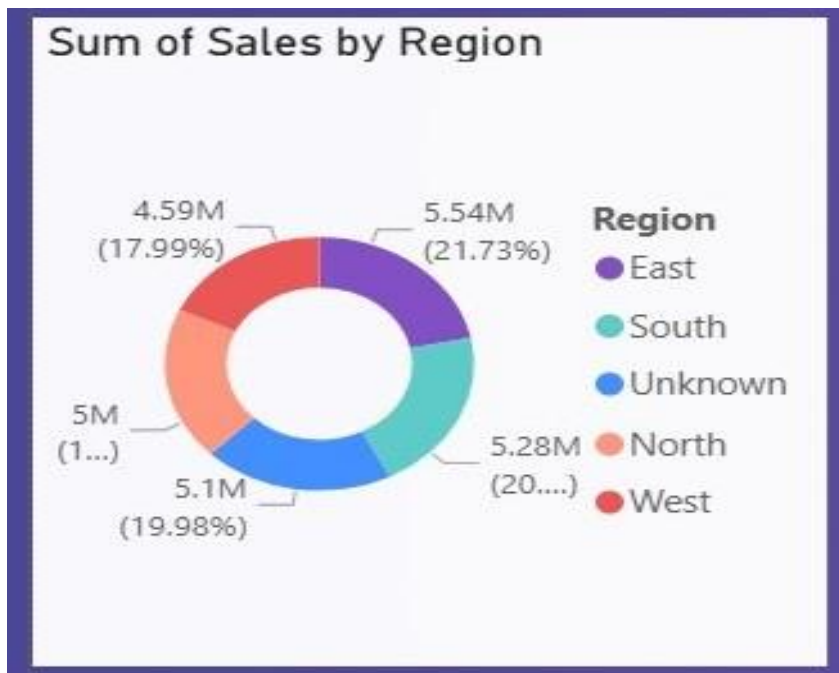
- Sum of Sales by Order Date and Order ID



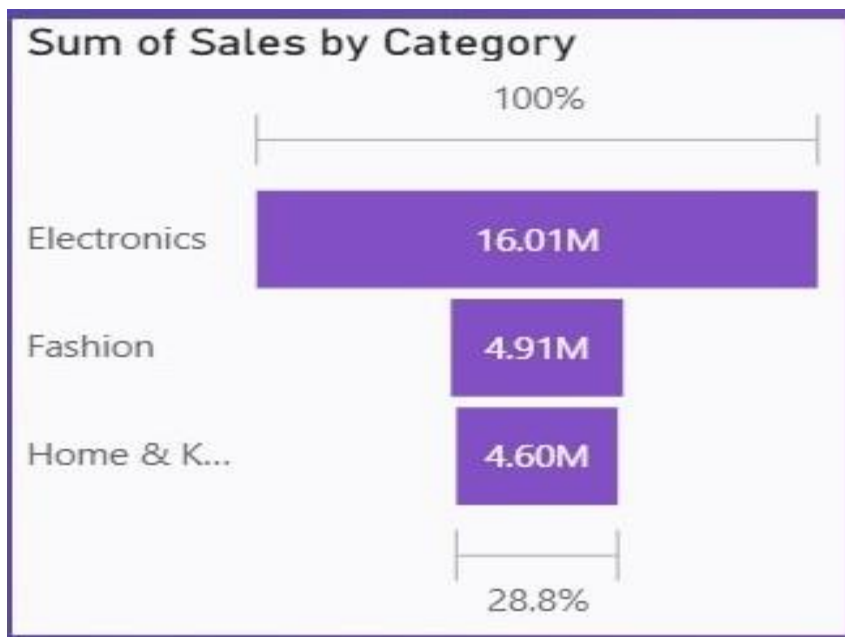
- Sum of Sales by Segment



- Sum of Sales by Region



- Sum of Sales by Category

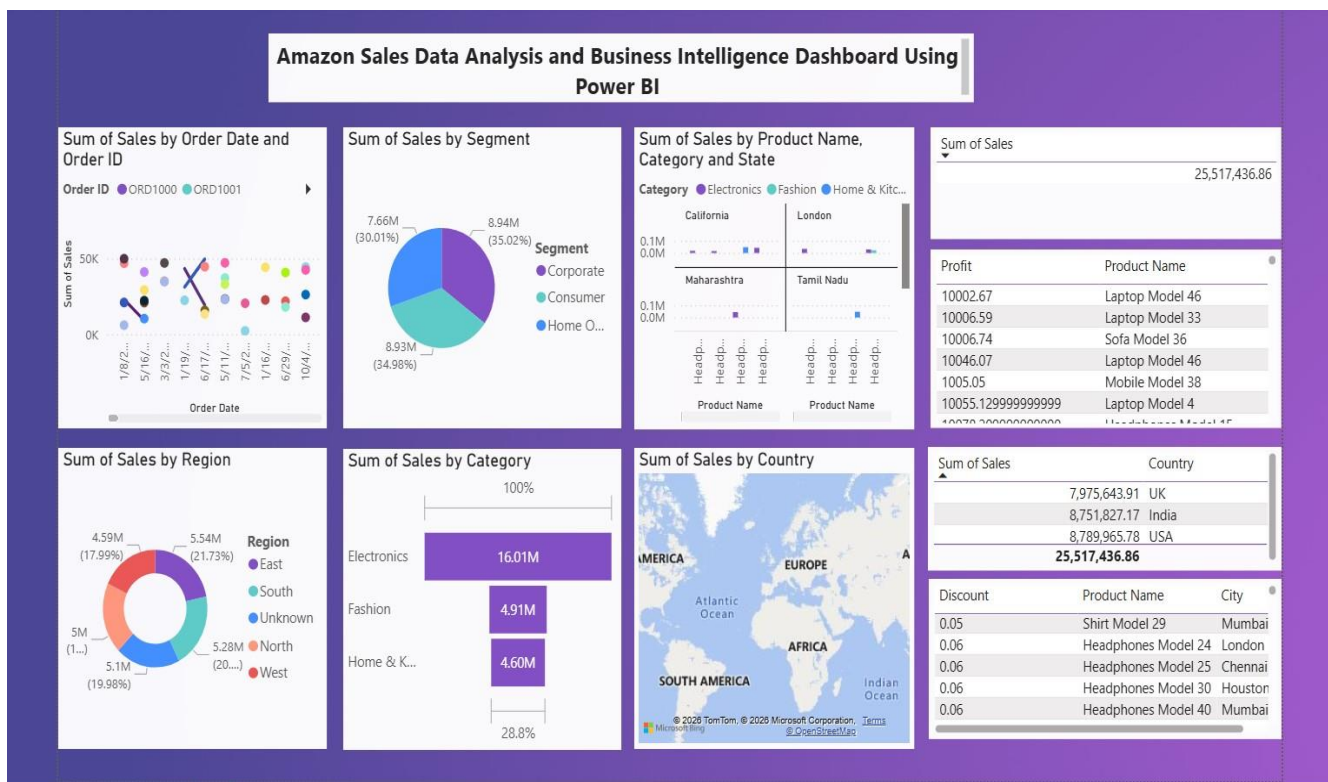


- Profit vs Discount Analysis

Discount	Product Name	City
0.05	Shirt Model 29	Mumbai
0.06	Headphones Model 24	London
0.06	Headphones Model 25	Chennai
0.06	Headphones Model 30	Houston
0.06	Headphones Model 40	Mumbai

Interactive Features

- Slicers for Region, Category, Segment, Date
- Drill-down and cross-filtering
- Dynamic visuals based on user selection



9. Insights & Conclusions

Descriptive Analysis (What Happened?)

- Certain regions recorded high sales volume but low profit because heavy discounts reduced overall margins.
- Technology and Office Supplies categories consistently generated stable and positive profits across periods.
- The Consumer segment placed the highest number of orders, contributing significantly to total revenue.
- Higher discount percentages are clearly associated with reduced profit margins.
- A small number of top-performing products contributed to the majority of total sales revenue

Diagnostic Analysis (Why Did It Happen?)

- Heavy promotional discounting in some regions increased sales volume but directly reduced profitability.
- Technology and Office Supplies likely have better pricing control and lower discount dependency.
- The Consumer segment dominates due to broader market demand and frequent repeat purchases.
- Profit margins shrink because discounts reduce unit contribution without proportional cost reduction.
- Revenue concentration occurs because high-demand products dominate customer preference.

Predictive Analysis (What Is Likely to Happen?)

- If heavy discounting continues, profit margins in high-sales regions may decline further.
- Technology and Office Supplies are expected to maintain steady profitability if pricing strategy remains stable.
- Consumer segment orders will likely continue to grow due to strong purchasing behavior.
- Increasing discount levels may further weaken overall profitability.

- Top-performing products will continue driving most revenue unless new competitive products emerge.

Prescriptive Analysis (What Should Be Done?)

- Optimize discount strategy in high-sales regions to balance revenue growth and profit margins.
- Focus marketing and inventory investment on Technology and Office Supplies categories.
- Develop loyalty programs targeting the Consumer segment to increase retention without heavy discounts.
- Implement discount caps to prevent excessive margin erosion.
- Diversify product promotion strategy to reduce dependency on a small number of high-revenue products.

10. Conclusion

This project successfully demonstrates how **Microsoft Excel** can be used for efficient data cleaning, transformation, and preparation before analysis. The structured dataset was then integrated into **Microsoft Power BI** to create an interactive and insightful business intelligence dashboard. Through proper data modeling and visualization techniques, complex sales data was transformed into meaningful insights. The dashboard monitor overall sales performance across regions, categories, and customer segments in real time. It helps identify key profit drivers and highlights areas where heavy discounts are impacting margins. The visual reports simplify trend analysis and comparison, supporting faster and more accurate interpretation of data. Decision-makers can easily detect high-performing products and underperforming regions. The integration of analytical techniques improves strategic planning and operational efficiency. By leveraging interactive filters and drill-down features, users can explore detailed performance metrics. Overall, this project proves how combining Excel and Power BI supports effective, data-driven business decision-making and enhances organizational performance.

11. Future Enhancements

- Add predictive sales forecasting
- Perform customer lifetime value analysis and Integrate SQL or Python for advanced analytics.